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 Studierichting: Informatica
 Oefeningenreeks 3H nr. 8.2

$$r = 2 \cos(2\theta)$$

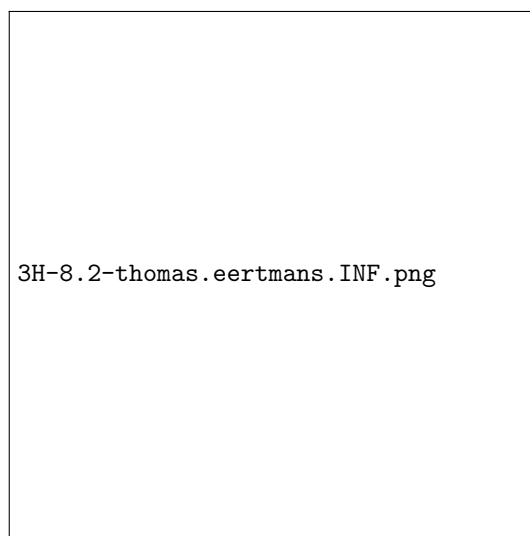
Periode = π , beperkt domein: $[0, \pi]$

$$\text{nulpunten: } 2 \cos(2\theta) = 0 \Leftrightarrow \cos(2\theta) = 0 \Leftrightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$r' = 2(-\sin 2\theta) \cdot 2 = 4(-\sin 2\theta)$$

$$\text{nulpunten: } 4(-\sin 2\theta) = 0 \Leftrightarrow \theta = 0, \frac{\pi}{2}, \pi$$

θ	0	$\frac{\pi}{4}$		$\frac{\pi}{2}$	$\frac{3\pi}{4}$		π	
$r(\theta)$	+	+	0	-	-	-	0	+
$r'(\theta)$	0	-	-	-	0	+	+	+
$r(\theta)$	$M(2)$	$\searrow$	$\searrow$	$\searrow$	$m(-2)$	$\nearrow$	$\nearrow$	$\nearrow$
								$M(2)$



References